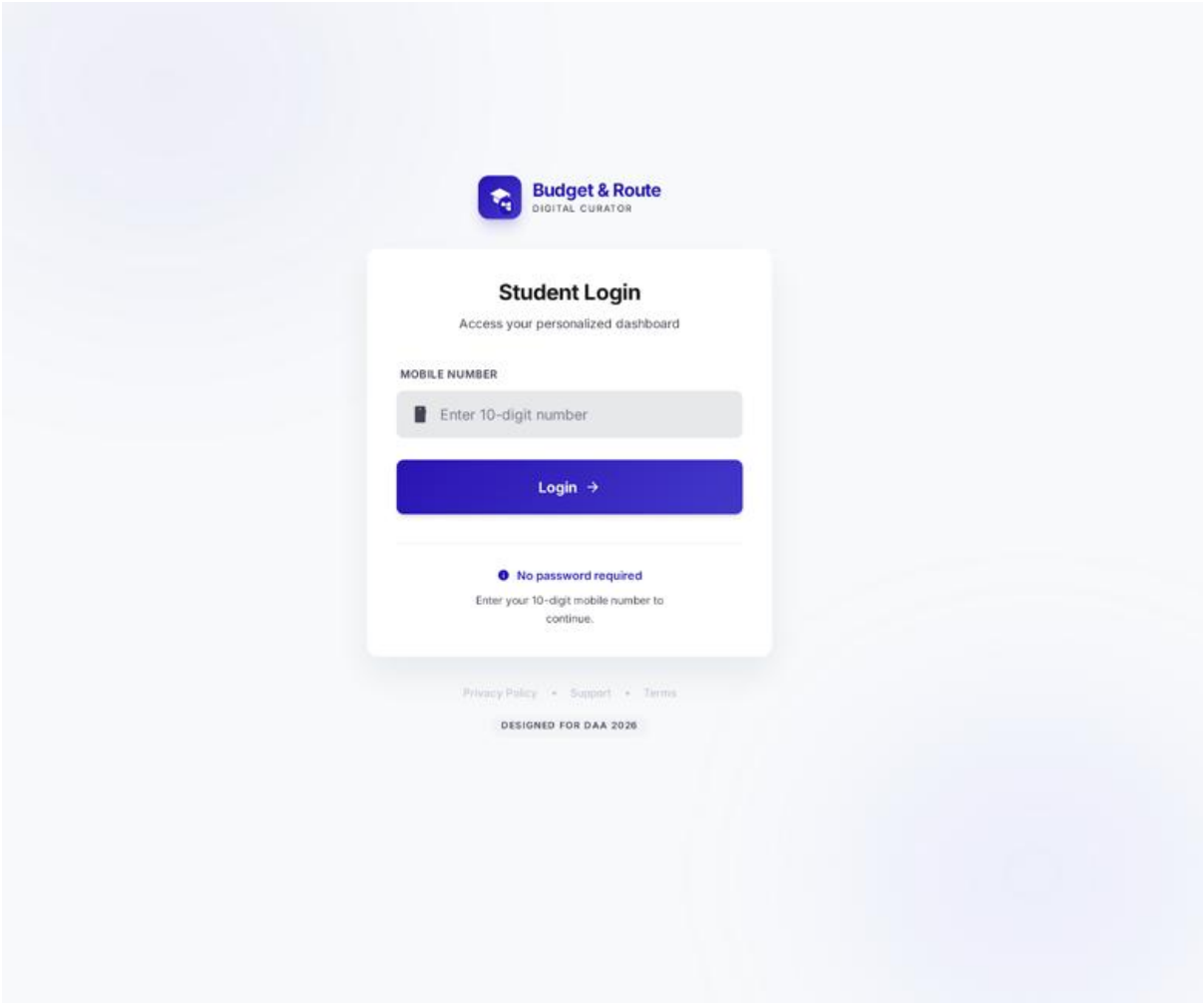
Budget & Route
DIGITAL CURATOR
Student Login
Access your personalized dashboard
MOBILE NUMBER
Enter 10-digit number
Login →
No password required
Enter your 10-digit mobile number to continue.
Privacy Policy • Support • Terms
DESIGNED FOR DAA 2026

Academic Curator | Dashboard | Analysis | Archive | Reports | Search algorithms...

CURATOR PRO
DEEP ANALYSIS MODE

- OVERVIEW
- ALGORITHMS
- DATA SETS
- COMPLEXITY
- SETTINGS

Welcome back, Student!

Your research dashboard is synchronized. Ready to explore new computational efficiencies?

OPTIMAL PRIORITY

Budget Planner

Optimize resource allocation using dynamic programming models for academic expenditures.

Go to Algorithm →

Route Finder

Shortest path analysis via Dijkstra's and A* protocols for campus mapping.

Go to Algorithm →

MARKET ANALYSIS

Top Products

Real-time ranking of curated academic tools and resources based on peer-reviewed utility scores.

JOURNAL A | IDE PRO | DATASETS | AI REF

Go to Algorithm

SYSTEMS ACTIVE
128.4 ms

QUEUE STATUS
Operational

Academic Curator / Research v4.2.0
Deep Analysis Mode Active

Budget & Route

NAVIGATION

- Overview
- Budget Analysis
- Route Finder
- Product Ranking

New Analysis

Settings

Documentation

KNAPSACK OPTIMIZER

Budget Allocation Engine

Maximize your total value within strict budget constraints using the 0/1 Knapsack dynamic programming algorithm.

CONFIGURATION

TOTAL BUDGET (W)

1200

QUICK ADD PRODUCT

Name

Price | Value

+ Add Item to List

Optimize Selection

TOTAL VALUE
1,480

TOTAL COST
1,150

REMAINING
50

OPTIMIZED INVENTORY SELECTION

0/1DECISIONMATRIX

PRODUCT NAME	PRICE	VALUE	DECISION
High-End Laptop	$800	900	SELECTED
Ergonomic Chair	$350	580	SELECTED
External Monitor	$250	400	REJECTED
Mechanical Keyboard	$100	150	REJECTED

SUBPROBLEM DP TABLE

C \ I	0	1	2	3	4
100	0	150	150	150	150
200	0	150	300	300	300

Cell dp[i][w] represents maximum value with items 1..i and capacity w.

METHOD COMPARISON

GREEDY RESULT 1,350

KNAPSACK RESULT (OPTIMAL) 1,480

OBSERVATION

The greedy approach fails here due to item indivisibility, capturing 9.6% less value than our DP model.

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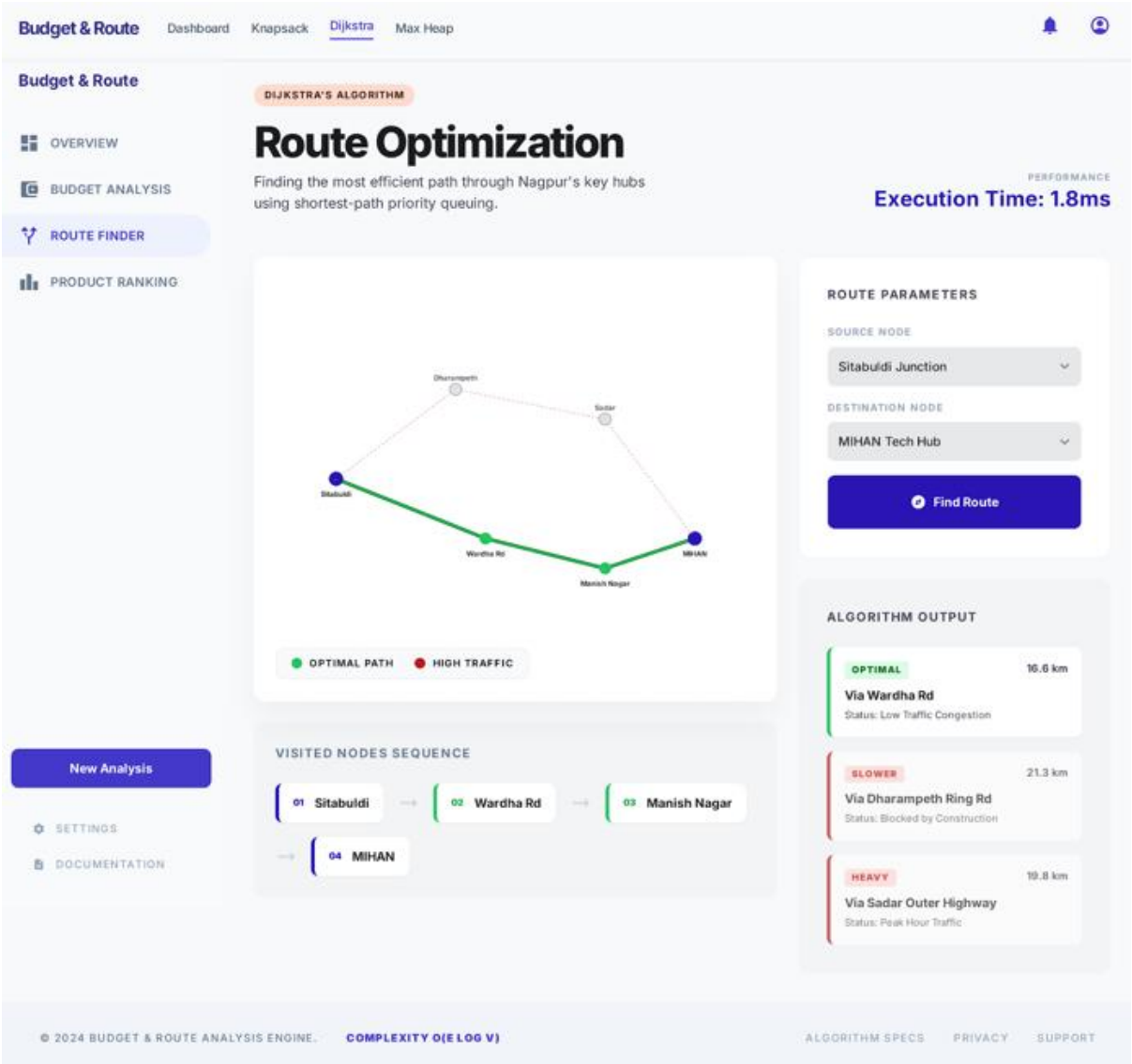
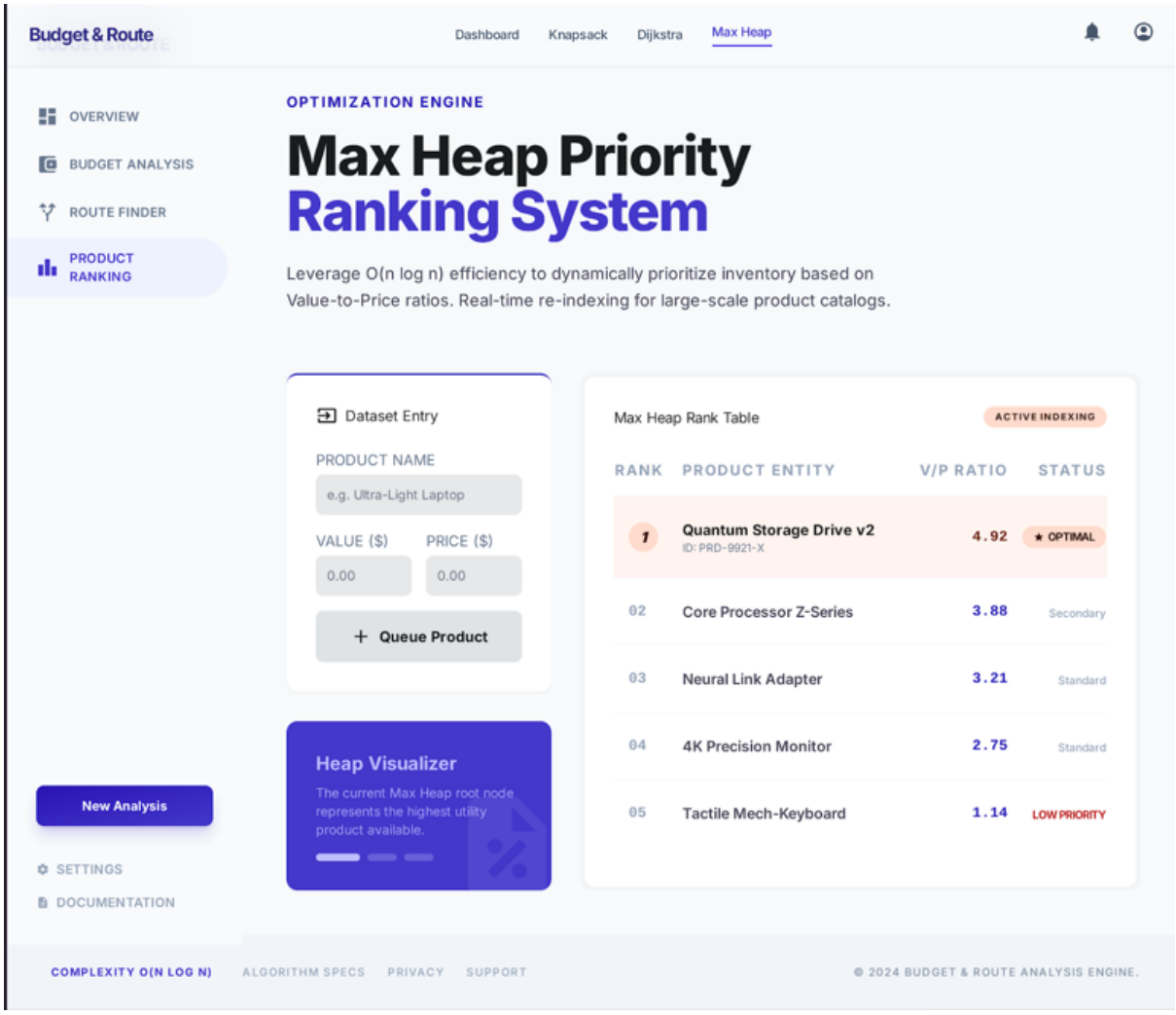

Budget & Route
Dashboard
Knapsack
Dijkstra
Max Heap
Budget & Route
OVERVIEW
BUDGET ANALYSIS
ROUTE FINDER
PRODUCT RANKING
New Analysis
SETTINGS
DOCUMENTATION
DIJKSTRA'S ALGORITHM
Route Optimization
Finding the most efficient path through Nagpur's key hubs using shortest-path priority queuing.
PERFORMANCE
Execution Time: 1.8ms
Dharampeth
Sadar
Sitabuldi
Wardha Rd
Manish Nagar
MIHAN
OPTIMAL PATH
HIGH TRAFFIC
VISITED NODES SEQUENCE
01 Sitabuldi
02 Wardha Rd
03 Manish Nagar
04 MIHAN
ROUTE PARAMETERS
SOURCE NODE
Sitabuldi Junction
DESTINATION NODE
MIHAN Tech Hub
Find Route
ALGORITHM OUTPUT
OPTIMAL
16.6 km
Via Wardha Rd
Status: Low Traffic Congestion
SLOWER
21.3 km
Via Dharampeth Ring Rd
Status: Blocked by Construction
HEAVY
19.8 km
Via Sadar Outer Highway
Status: Peak Hour Traffic
© 2024 BUDGET & ROUTE ANALYSIS ENGINE.
COMPLEXITY O(E LOG V)
ALGORITHM SPECS
PRIVACY
SUPPORT


Budget & Route
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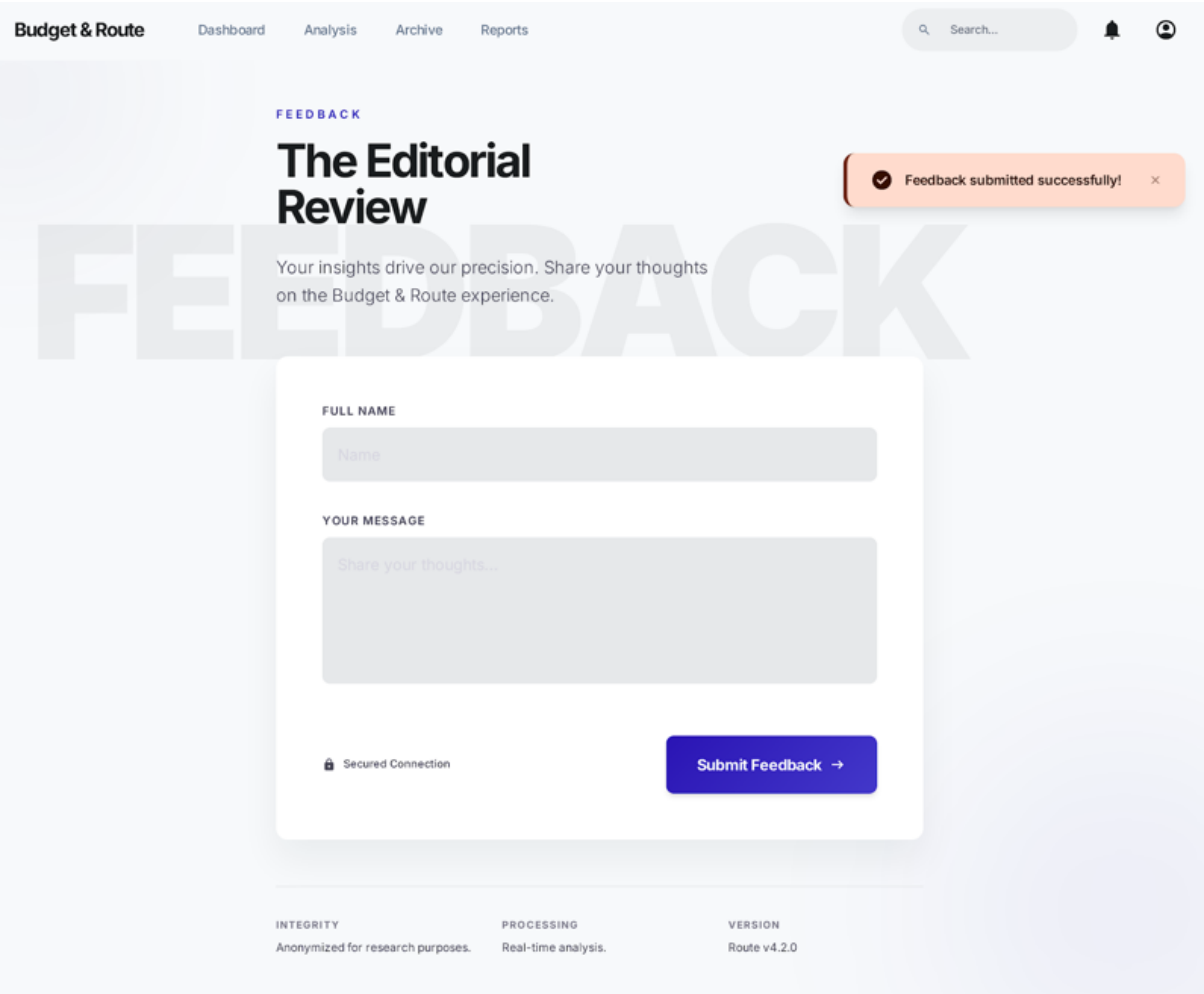
ROUTE FINDER
PRODUCT RANKING
New Analysis
SETTINGS
DOCUMENTATION
OPTIMIZATION ENGINE
Max Heap Priority Ranking System
Leverage O(n log n) efficiency to dynamically prioritize inventory based on Value-to-Price ratios. Real-time re-indexing for large-scale product catalogs.
Dataset Entry
PRODUCT NAME
e.g. Ultra-Light Laptop
VALUE ($)
PRICE ($)
0.00
0.00
Queue Product
Heap Visualizer
The current Max Heap root node represents the highest utility product available.
Max Heap Rank Table
ACTIVE INDEXING
RANK
PRODUCT ENTITY
V/P RATIO
STATUS
1
Quantum Storage Drive v2
ID: PRD-9921-X
4.92
OPTIMAL
02
Core Processor Z-Series
3.88
Secondary
03
Neural Link Adapter
3.21
Standard
04
4K Precision Monitor
2.75
Standard
05
Tactile Mech-Keyboard
1.14
LOW PRIORITY
COMPLEXITY O(N LOG N)
ALGORITHM SPECS
PRIVACY
SUPPORT
© 2024 BUDGET & ROUTE ANALYSIS ENGINE.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Dashboard
Analysis
Archive
Reports
Search...
FEEDBACK
The Editorial Review
Feedback submitted successfully!
FEEDBACK
Your insights drive our precision. Share your thoughts on the Budget & Route experience.
FULL NAME
Name
YOUR MESSAGE
Share your thoughts...
Secured Connection
Submit Feedback →
INTEGRITY
Anonymized for research purposes.
PROCESSING
Real-time analysis.
VERSION
Route v4.2.0